

FANUC Robot **series**

R-30iB and R-30iB Plus CONTROLLER

Dynamic Path Modification

User's Guide

MARUBDPMO02141E REV B

Version 8.30 and later

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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FANUC America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

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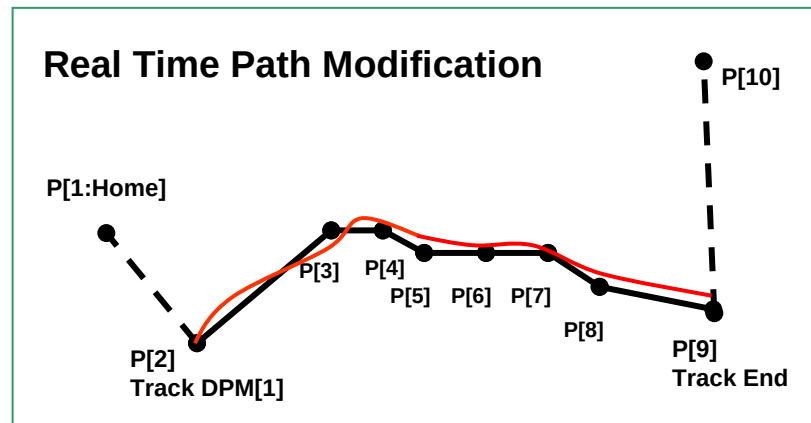
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1. Dynamic Path Modification Overview

DPM provides dynamic path modification with flexible third party sensor interface. It supports both modal DPM instruction and inline DPM instruction. Modal DPM is designed for the application that requires real time path modification based on the sensor information along the entire motion path. The inline DPM is designed for the application that requires real time modification on the destination position for each motion segment.

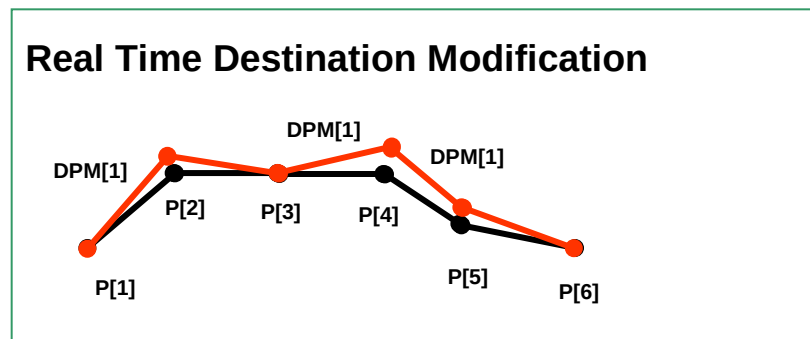
1.1 Modal DPM

- Real-time path modification
- Modal Track DPM instruction
- Continuous path modification
- Real time sensor offset (x,y,z,w,p,r)
- Flexible sensor interface (Digital I/O, Analog I/O, Group I/O and system variables)
- Path update rate at 8ms
- Quick path correction response (offset after JBF)



1.2 Inline DPM

- Real-time path destination modification
- Inline DPM instruction for any motion line
- Third party sensor for path segment destination offset (x,y,z,w,p,r) during run time
- Smooth Segment Path Transition
- Single Step and constant path at different speed override



2. Assumptions and Limitations

- DPM supports multi-group motion but limits to two DPM control groups per controller
- DPM supports weave operation for modal DPM motion
- DPM is independent of the PROCESS and can be applied to ALL applications
- DPM supports extended axis (integral rail configuration) and Topload
- DPM can co-exist but can't work simultaneously with any tracking related operation such as
 - o TAST/AVC
 - o Servo-Robot Tracking
 - o Line-Tracking
- DPM can support orientation control but it is limited to UFRAM and UTOOL relative and with CP_MODE3 resume mode.

3. DPM Installation Procedure

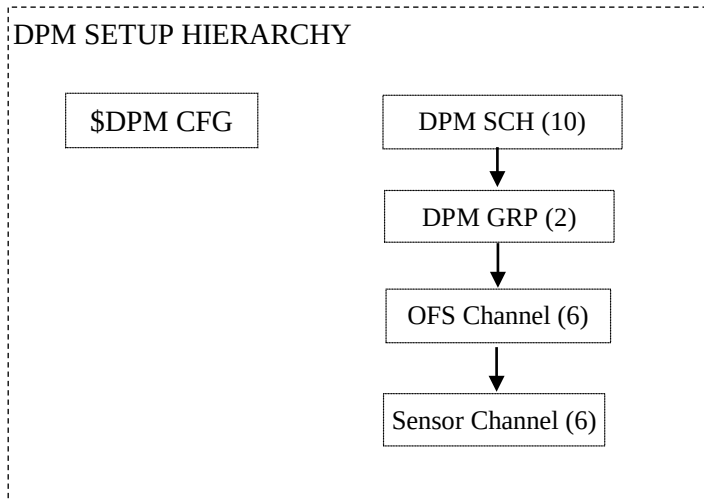
DPM option requires load and enable of the constant path option.

- Load R739 option in initial start
- DPM requires the Constant Path Option to be loaded and enabled.

4. DPM SETUP

- DPM Schedule Setup
- DPM Group Setup
- DPM Channel Setup

The block diagram shows the hierarchy of the DPM setup.



Following sections give the detailed descriptions of DPM UIF setup.

4.1 DPM UIF Configuration Setup

Follow procedures below to get DPM Configuration setup menu

- Press MENU.
- Select SETUP.
- Press F1, [TYPE].
- Select DPM SETUP.
- Press F3, CONFIG, You will see a screen similar to the following

```

DPM CFG SETUP
1234567890123456789012345678901234
                                     2/6

  1 DPM function                      ENABLED
  2 Instruction type:                  MODAL
  3 Offset BEF/AFT filter:            AFTER
  4 Orientation CTRL                   DISABLED
  5 Delay time (inline)                5 ITP
  6 Motion group mask: [1,1,*,*,*,*,*]

[ TYPE ]    SCHEDULE    INLINE    MODAL
  
```

4.1.1 DPM UIF Configuration Descriptions

Item	Item Description
DPM function	ENABLED or DISABLED for the function.
Instruction type	MODAL or INLINE for the operation.
Offset BEF/AFT filter	BEFORE: offset is applied before filter. AFTER: offset is applied after filter. Require re-cycle power for change to be effective.
Orientation CTRL	Enabled: Enable orientation control, Disabled: disable orientation control.
Delay time (inline DPM)	Specifies window time (ITP delay) for INLINE DPM to accept offset after a new segment starts.
Motion group mask: [1,1,*,*,*,*,*]	1 st 1 of the motion group maps to DPM group 1 and 2 nd 1 maps to DPM group 2. Require re-power of controller for change to be effective.

4.2 DPM List Schedule Setup

Follow the procedures below to get DPM list schedule menu

- Press MENU.
- Select SETUP.
- Press F1, [TYPE].
- Select DPM SETUP.

You will see a screen similar to the following

DATA DPM Sched				
12345678901234567890123456789012345678901234				
Schedule	ENABLED	COMMENTS	1/10	
1	TRUE	Schedule 1		
2	TRUE	Schedule 2		
3	TRUE	Schedule 3		
4	TRUE	Schedule 4		
5	TRUE	Schedule 5		
6	TRUE	Schedule 6		
7	TRUE	Schedule 7		
8	TRUE	Schedule 8		
9	TRUE	Schedule 9		
10	TRUE	Schedule 10		
[TYPE]	DETAIL	CONFIG	TRUE	FALSE

Set TRUE/FALSE to enable/disable this schedule.

4.3 DPM Detail Schedule Setup

Select DETAIL (F2) from the list schedule menu, you will see detail schedule menu. The detail schedule menu consists of dpm group and dpm channel setup. The group and channel display items are further dependent on DPM type and channel input type.

4.3.1 MODAL Detail Screen for SYSVAR Input Type

DPM DET SCH	
12345678901234567890123456789012345678901234	
SCHD 1, DPMGRP 1, CHAN 1 11/12	
1 DPM type:	MODAL
2 Offset ref. frame:	UFRAME
3 Offset accumulate:	FALSE
4 Stationary Track:	NO
5 Station Track sych:	DI[0]
6 Channel enabled:	TRUE
7 On-The-Fly Input:	DI[0]
8 Max offset/path:	10.00 mm
9 Max offset/ITP:	.20 mm
10 Min offset/ITP:	00.00 mm
11 Offset input type	SYSVAR
12 Offset value	0.00 mm
SCHEDULE DPMGRP CHANNEL [CHOICE] HELP	

4.3.2 MODAL Detail Screen for Digital Input Type

DPM DET SCH	
12345678901234567890123456789012345678901234	
SCHD 1, DPMGRP 1, CHAN 1 1/14	
1 DPM type:	MODAL
2 Offset ref. frame:	UFRAME
3 Offset accumulate:	FALSE
4 Stationary Track:	NO
5 Station Trck sych:	DI[0]
6 Channel enabled:	TRUE
7 On-The-Fly Input:	DI[0]
8 Max offset/path:	10.00 mm
9 Max offset/ITP:	0.20 mm
10 Min offset/ITP:	0.00 mm
11 Offset input type:	DIGITAL
12 Plus bump input:	DI[0]
13 Minus bump input:	DI[0]
14 Bump value/ITP:	0.05 mm
SCHEDULE DPMGRP CHANNEL [CHOICE] HELP	

4.3.3 MODAL Detail Screen for Group Input Type

DPM DET SCH			
12345678901234567890123456789012345678901234			
SCHD 1, DPMGRP 1, CHAN 1		11/13	
1	DPM type:	MODAL	
2	Offset ref. frame:	UFRAME	
3	Offset accumulate:	FALSE	
4	Stationary Track:	NO	
5	Station Trck sych:	DI[0]	
6	Channel enabled:	TRUE	
7	On-The-Fly Input:	DI[0]	
8	Max offset/path:	10.00	mm
9	Max offset/ITP:	0.20	mm
10	Min offset/ITP:	0.00	mm
11	Offset input type:	GROUP	
12	Group Index:	GI[10]	
13	Scale factor:	1000.00	
SCHEDULE DPMGRP CHANNEL [CHOICE] HELP			

4.3.4 MODAL Detail Screen for Analog Input Type

DPM DET SCH			
12345678901234567890123456789012345678901234			
SCHD 1, DPMGRP 1, CHAN 1		11/24	
1	DPM type:	MODAL	
2	Offset ref. frame:	UFRAME	
3	Offset accumulate:	FALSE	
4	Stationary Track:	NO	
5	Station Trck sych:	DI[0]	
6	Channel enabled:	TRUE	
7	On-The-Fly Input:	DI[0]	
8	Max offset/path:	10.00	mm
9	Max offset/ITP:	0.20	mm
10	Min offset/ITP:	0.00	mm
11	Offset input type:	ANALOG	
12	Analog input port:	AI[0]	
13	A/D conversion slope	1.0	
14	A/D conversion offset	0	
15	Position CTRL Gain:	10.0	
16	Reference mode:	FEEDBK	
17	Reference value:	0	
18	Start feedback number:	0	
19	Moving average number:	10	
20	DPM Starts after:	10 ITP	
21	Min. threshold value:	0	
22	Max. threshold value:	100	
23	Calibration is done:	TRUE	
24	Ring buffer size	128	
SCHEDULE DPMGRP CHANNEL [CHOICE] HELP			

4.3.5 MODAL Detail Schedule Descriptions

	Items	Description
	DPM type	Set MOAL for this DPM Group
Group	Offset ref. frame	1: PATH. Limited to positional offset only, offset channels 1, 2 and 3 corresponding to x, y and z w/r PATH. 2: TOOL-PATH: Limited to positional offset only, offset channels 1, 2 and 3 corresponding to x, y and z w/r TOOL-PATH. 3: UFRAME: offset is relative to current user frame. Offset channels 1, 2, 3, 4, 5 and 6 correspond to x, y, z, w, p, r w/r UFRAME. 4: UTOOL: offset is relative to current tool frame. Offset channels 1, 2, 3, 4, 5 and 6 correspond to x, y, z, w, p, r w/r UTOOL.
	Offset accumulate	- True for Accumulated: if a new offset is commanded while old offset is in progress the new offset would be added to the old offset to establish a new total offset - FALSE for Non-Accumulated: if a new offset is commanded while the old offset is in progress the new offset would become the new total offset and un-consumed old offset will be flushed.
	Stationary Track	- Yes to enable stationary track. - No for regular track mode.
	Station Track sych DI[idx]	Stationary track digital index used to terminate the stationary tracking: - DI[sync_di] = OFF, DPM will perform stationary track. - DI[sych_di] = ON will terminate stationary track for this motion line and continues to next motion line. If next motion line is also a stationary track, you should reset DI[sync_di] = OFF prior to the motion end.
Offset Channel	Channel enabled	Enable or disable this channel.
	On-The-Fly input	Input I/O index for feedback.
	Max offset/path	Allowed total path deviation in mm.
	Max offset/ITP	Allowed max absolute path change in mm per ITP.
	Min offset/ITP	Minimum offset per ITP before control take effect in mm.
	Channel input type	ANALOG: offset is derived from an analog signal. GROUP: offset is derived from group input. DIGITAL: offset is derived from Bump Digital Input. SYSVAR : User set offset via sysvars.

Analog Input Channel	Analog input port AI[idx]	Analog port number.
	A/D conversion slope	Analog A/D calibration parameter.
	A/D conversion offset	Analog A/D calibration parameter.
	Position ctrl gain	Positional control gain for the PID control.
	Reference mode	▪ MASTER: user set reference value. ▪ FEEDBK: regular feedback mode. ▪ FEEDBK1: special feedback mode.
	Reference value	Reference value used to generate offset.
	Start feedback number	Specify first no. of samples to skip before calculating reference value. Only used in FEEDBK1 mode.
	Moving average number	Average feedback number.
	DPM start after	Specifies the number of ITP before DPM control.
	Min. threshold value	Minimum calibrated feedback value. Data will be marked as invalid when data is less than this value.
	Max. threshold value	Maximum calibrated feedback value. Data will be marked as invalid when data is greater than this value.
	Calibration is done	Set TRUE when calibration is done.
	Ring buffer size	Ring buffer size. MUST be larger than moving average number.
Group Input Channel	Group index GI[idx]	Index for the group input.
	Scale factor	Scale factor for the group input.
Digital Input Channel	Plus bump input DI[idx1]	Index for bump DIN for plus direction.
	Minus bump input DI[idx2]	Index for bump DIN for minus direction.
	Bump value/ITP	Bump offset per ITP when bump DIN is on.
Sysvar Input Channel	Offset value	Channel offset value.

4.3.6 INLINE Detail Schedule Setup

For INLINE DPM, the offset is applied to destination position. The offset is normally provided at one time for each motion segment. Therefore, the sensor channel setup is not displayed.

DPM DET SCH	
1234567890123456789012345678901234	
SCHD 1, DPMGRP 1, CHAN 1 1/9	
1 DPM type:	INLINE
2 Offset ref. frame:	UFRAME
3 Offset accumulate:	FALSE
4 Group input type:	SINGL-PR
5 Position Register:	REG[0]
6 Channel enabled:	TRUE
7 Max offset/path:	10.00 mm
8 Max offset/ITP:	0.20 mm
9 Min offset/ITP:	0.00 mm
SCHEDULE DPMGRP CHANNEL [CHOICE] HELP	

4.3.7 INLINE Detail Schedule Descriptions

	Items	Description
Group	DPM type	Set INLINE for this DPM Group
	Offset ref. frame	Same as MODAL DPM case
	Offset accumulate	- True for Accumulated: The offset stored in position register will not be flushed after being used. - FALSE for Non-Accumulated: The offset stored in position register will be flushed after being used.
	Group input type	- USER-DEF: User defined offset. - SINGLE-PR: Offset stored at specified position register for all DPM move. - MULTI-PR: Offset stored at different position register where the register index must match the DPM move line number.
	Position Register REG[idx]	Specify the position register for SINGLE-PR input type
	Channel enabled	Same as MODAL DPM case
Offset Channel	Max offset/path	Same as MODAL DPM case
	Max offset/ITP	Same as MODAL DPM case
	Min offset/ITP	Same as MODAL DPM case

5. DPM Setup and DPM Control

Following sections give more detailed descriptions of the setup via dpm sysvars and its related dpm control.

5.1 DPM Configuration Setup (\$dpm_cfg)

5.1.1 DPM function (\$enable)

Allows you enable or disable DPM function

5.1.2 Motion group setup (\$grp_msk)

DPM supports two dpm motion groups per controller and you can map robot motion group to dpm group. For example,

- Motion group mask: [1,1,*,*,*,*,*] (\$grp_msk = 3) will map motion group 1 and 2 to dpm group 1 and 2 respectively.
- Motion group mask: [*,1,1,*,*,*,*] (\$grp_msk = 6) will map motion group 2 and 3 to dpm group 1 and 2 respectively.
- Re-power is required for the change to be effective.

5.1.3 Instruction type (\$dpm_inline)

- MODAL for real time path offset (\$dpm_inline = FALSE)
- INLINE for real time destination offset (\$dpm_inline = TRUE)

DPM supports MODAL and INLINE DPM instructions. You can switch the instruction type at any time without re-powering the controller.

5.2 DPM Schedule Setup (\$dpm_sch[sch])

DPM can support up to 10 schedules and each schedule can support up to two dpm groups.

5.2.1 DPM TYPE (\$dpm_type)

- INLINE : for real time destination offset (inline DPM application)
- MODAL: for real time path offset (modal DPM application)

You can specify INLINE or MODAL DPM type to any dpm group in the schedule. However, you can only execute DPM motion whose dpm_type is the same as the Instruction type in dpm configuration setup. Otherwise, you will get DPMO-027 Mismatch DPM type alarm.

5.2.2 Offset ref. frame (\$ofs_fram)

DPM supports following four offset reference frames:

- 1: PATH: the x direction is the path direction, the y direction is perpendicular to both x and UFRAME z and the z direction is perpendicular to both x and z. Path relative is limited to positional offset only, offset channels 1, 2 and 3 corresponding to x, y and z w/r PATH.
- 2: TOOL-PATH: the x direction is the path direction, the y direction is perpendicular to both x and TOOL z and the z direction is perpendicular to both x and z. TOOL-Path relative is limited to positional offset only, offset channels 1, 2 and 3 corresponding to x, y and z w/r TOOL-PATH.
- 3: UFRAME: the offset is relative to the current user frame. Offset channels 1, 2, 3, 4, 5 and 6 correspond to x, y, z, w, p, r w/r UFRAME.

- 4: UTOOL: the offset is relative to the current tool frame. Offset channels 1, 2, 3, 4, 5 and 6 correspond to x, y, z, w, p, r w/r UTOOL.

Warning: On the fly change of the \$ofs_fram is prohibited in one DPM motion segment. However, user can change \$ofs_fram by using different track schedule (see detail in programming section)

5.2.3 Offset accumulate (\$ofs_accu)

DPM supports accumulated and Non-accumulated offset modes for Modal and Inline DPM

- Modal DPM type
 - True for Accumulated: if a new offset is commanded while the old offset is in progress the new offset would be added to the old offset to establish a new total offset.
 - FALSE for Non-Accumulated: if a new offset is commanded while the old offset is in progress the new offset would become the new total offset and un-consumed old offset will be flushed.
- INLINE DPM type
 - True for Accumulated: The offset stored in position register will not be flushed after being used.
 - FALSE for Non-Accumulated: The offset stored in position register will be flushed after being used.

5.2.4 Stationary Tracking (\$strk_mode and \$sync_di)

DPM supports stationary track motion for modal DPM with FINE term motion. In the stationary track motion, the robot will move to a taught position with or without any offset and to continue to follow the offset input (via joystick command or other sensors input in six channels) in Cartesian move relative to UTOOL or USER frame. The stationary track for this motion will not end until a specified synchronization digital input is set to TRUE. Then, robot will continue to move to next motion segment. You can use following sysvars for stationary tracking control

- \$strk_mode
 - o 0: for regular track mode
 - o 1: enable stationary tracking for this group
- \$sync_di: stationary track digital index used to terminate the stationary tracking
 - o DI[sync_di] = OFF, DPM will perform stationary tracking
 - o DI[sych_di] = ON will terminate stationary track for this motion line and continues to next motion line. If next motion line is also a stationary track, you should reset DI[sync_di] = OFF prior to the motion end.

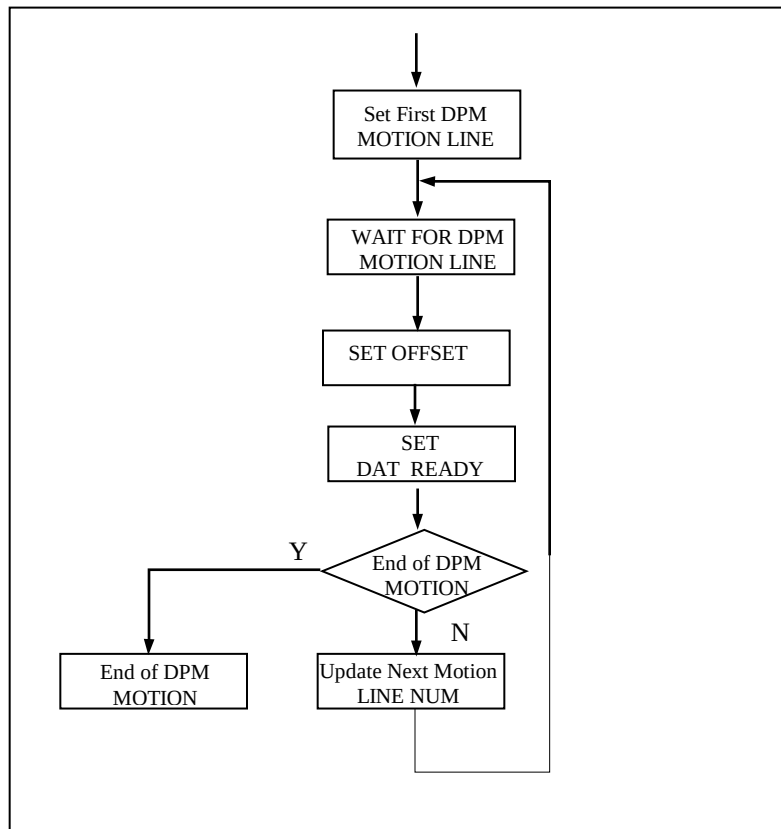
5.2.5 Group Input Type (\$ofs_input)

This item is only used by inline dpm. You can input the inline DPM offset by a single position register or multiple position registers.

- SINGLE-PR: Offset stored in one position register for all DPM move. You must set the position register index. The offset stored in the register will not be flushed after being used for accumulated offset mode and will be flushed after being used for non-accumulated mode.
- MULTI-PR: Offset stored in multiple position registers where the position register index must match to the DPM move line number.
- USER-DEF: User defined offset. This input type requires handshaking. User has to provide destination offset per each DPM motion line at the given time window. Following dpm group variables can be used for the handshaking between DPM and application program (such as KARAL)

- o \$DPM_SCH[sch].\$GRP[g].\$int_line: represents the current DPM motion line number and is updated by DPM at the beginning of each dpm motion line. The application program references this value to handshake to DPM task.
- o \$DPM_SCH[sch].\$GRP[g].\$dat_rdy: this is the handshaking flag. The handshaking sequence follows:
 - DPM initializes this value to zero at beginning of each dpm motion line
 - User application program sets this value to one after offset is set to \$ini_ofs
 - DPM will apply this offset to current DPM motion line if the offset is available within the segment window time (see delay time in DPM CFG SETUP)

Below illustrates the logic flow for the application programming



5.2.6 OFS Channel Setup \$ofs[ch]

Each robot group supports six offset channels, first three (1, 2 and 3) channels correspond to Cartesian X, Y and Z and next three channel (4, 5, and 6) correspond to W, P, and R for DPM motion control. The offset channel specifies DPM control parameter as well as the source of the sensor input. Each channel can be set independently.

- `$enable` Channel enable flag
- `$otf_di_idx` On The Fly digital input index
- `$chn_type` Specify offset channel input type
- `$ini_ofs` Channel offset input
- `$rem_ofs` Un-consumed channel offset (reference only)
- `$app_ofs` total applied channel offset (reference only)
- `$max_lim` allowed total path deviation in mm (path)
- `$max_inc` allowed max absolute path change in mm per ITP
- `$min_inc` threshold for the absolute offset to be considered
- `$a1` constant for 1st stage exponential filter
- `$a2` constant for 2nd stage exponential filter

5.2.7 Channel type \$ofs[ch].\$chn_type

DPM allows you to select offset input type independently among the channels in one group. For example, you can select bump DIN for lateral offset channel while selecting analog input signal for the vertical channel. In general, DPM supports following four input types for each offset channel:

- **AI_TYPE** (`$chn_type = 1`): Indirect offset that is derived from the sensor's raw signal. The raw signal is normally an analog signal (voltage or current) and is converted to path offset via a specified calibration and control algorithm. Indirect offset can support up to 3 analog channels.
- **GI_TYPE** (`$chn_type = 2`): The offset are scaled to a set of the integers to transmit to the controller via group digital input.
- **BI_TYPE** (`$chn_type = 3`): The offset is derived from the Bump Digital Input. Each bump input corresponds to a directional offset in a specified path direction. When a bump input is on, a user defined incremental offset is generated automatically. Bump digital input can support up to 12 channels ($\pm X$, $\pm Y$, $\pm Z$, $\pm W$, $\pm P$, $\pm R$).
- **SV_TYPE** (`$chn_type = 4`): This input type provides an easy interface when application using KAREL Ethernet socket messaging to obtain the offset from the 3rd party sensor device in customized protocol. The KAREL application can set the offset to the designated system variables directly for DPM control.

5.2.7.1 Analog input setup

5.2.7.1.1 AI Channel Setup \$ai_ch[ch]

- `$p_gain` P-Gain for the control
- `$str_cnt` Control start cycle count
- `$skp_cnt` first no of feedback samples to be skipped
- `$ave_num` No of samples for moving average
- `$fbk_mode` Master or feedback reference mode
- `$mst_done` System sets it true when reference value is set

- \$ref_val Reference value
- \$raw_fbk Raw feedback signal
- \$cal_fbk Calibrated feedback signal
- \$port_num Analog port number
- \$cal_done Set True when port is calibrated
- \$slope Port calibration parameter
- \$intercept Port calibration parameter
- \$buf_size Channel Ring buffer size to hold feedback data
- \$min_val Minimum calibrated feedback signal
- \$max_val Maximum calibrated feedback signal
- \$ctl_stage Adaptive stage control parameter
- \$param1_f Adaptive stage control parameter
- \$param2_f Adaptive stage control parameter

5.2.7.1.2 Analog Port Calibration

You must determine the intercept and slope based on A/D conversion and sensor analog signal range. Set these values to the channel system variable \$intercept and \$slope, and set the \$cal_done = TRUE.

5.2.7.1.3 \$fbk_mode.

Analog channel control supports three reference modes.

- o **MASTER** (\$fbk_mode = 1), the user set the reference value to \$ref_val.
- o **FEEDBK** (\$fbk_mode = 2), the reference value is set by system to \$ref_val based on first n samples where n = \$ave_cnt, that is

$$\text{\$ref_val} = \{\text{ave}(\text{fbk_val}) - \text{intercept}\} / \text{slope}$$

- o **FEEDBK1** (\$fbk_mode = 3), this mode allows you to apply additional constrains to calculate the reference value.
 1. You can specify a non-zero value to \$skp_cnt to skip the first \$skp_cnt feedback samples
 2. The consecutive number of samples (\$ave_cnt) must be within low (\$min_val) and upper (\$max_val) bound
 3. Once the above constrains are satisfied the reference value will be calculated as

$$\text{\$ref_val} = \{\text{ave}(\text{fbk_val}) - \text{intercept}\} / \text{slope}$$

The \$mst_done will be set to true automatically when ref_value is set and DPM tracking starts.

5.2.7.1.4 Analog proportional control

In analog proportional control, each channel offset is derived from the analog signal relative to a reference signal every ITP based on following equation

$$\text{offset} = \text{p_gain} * (\text{cal_fbk} - \text{ref_val})$$

The port_value is the moving average of the feedback data and the calibrated feedback value after A/D conversion is calculated as below

$$\text{cal_fbk} = (\text{port_value} - \text{intercept}) / \text{slope}$$

5.2.7.1.5 Analog adaptive stage control

The adaptive stage control dynamically adjusts ITP incremental offset threshold (\$max_inc) based on the deviation of the analog value from the reference value. It can effectively affect the tracking speed. The adaptive stage control uses three parameters

- \$ctl_stage stage threshold. Set value > 0 will enable this control
- \$parm1_f stage control rate
- \$parm2_f upper bound of max offset per ITP

```

deta1 = ABS {cal_fbk - ref_val}
if (delta > $ctl_stage) {
  stage = delta / $ctl_stage
  $max_inc = stage * $parm1_f
  If ($max_inc > $parm2_f)
    $max_inc = $parm2_f
  If ($max_inc < -$parm2_f)
    $max_inc = -$parm2_f
}

```

5.2.7.2 Group I/O input setup

- \$dpm_sch[sch].\$grp[grp].\$gi_ch[ch].\$di_idx group I/O index
- \$dpm_sch[sch].\$grp[grp].\$gi_ch[ch].\$scale scale factor for group input

The DPM will scan group i/o every ITP and determine the channel offset as

$$\text{offset} = \text{gi_value} / \text{scale}$$

5.2.7.3 Bump I/O input setup

- \$dpm_sch[sch].\$grp[grp].\$gi_ch[ch].\$di_idx1 Plus direction for the channel
- \$dpm_sch[sch].\$grp[grp].\$gi_ch[ch].\$di_idx2 Minus direction for the channel
- \$dpm_sch[sch].\$grp[grp].\$gi_ch[ch].\$bump_ofs incremental bump offset in mm
- \$dpm_sch[sch].\$grp[grp].\$gi_ch[ch].\$bump_rate bump offset rate

The DPM will scan bump DIN every ITP and determine the channel offset as

$$\text{offset} = + \text{bump_rate} * \text{bump_ofs} \quad \text{if DI[di_idx1]} = \text{ON}$$

$$\text{offset} = - \text{bump_rate} * \text{bump_ofs} \quad \text{if DI[di_idx2]} = \text{ON}$$

When plus and minus bump input for the same channel are ON at the same time, the plus offset will be produced.

5.2.7.4 System variable input

In the system variable input, user application program (such as KAREL) can directly set the channel offset to the system variables.

- `$dpm_sch[sch].$grp[grp].$ofs[ch].$ini_ofs`

DPM will scan the `$ini_ofs` for each channel offset every ITP and clear `$ini_ofs` once the offset is being used.

5.2.8 On-The-Fly DPM Control

When DPM is dynamically disabled, DPM will flush any un-consumed offset and ignore any new offset until the DPM is dynamically enabled again. There are two ways you can OTF enable or disable the DPM motion for each channel:

- Toggle system variable in the schedule `$dpm_sch[sch_no].$grp[1].$ofs[ch].$enable`
- Toggle the stage of the DIN to enable (DIN = OFF) or disable (DIN = ON) the DPM motion.
- The DIN is specified by `$dpm_sch[sch_no].$grp[1].$ofs[ch].$otf_di_idx`

6. Summary for DPM System Variables

6.1 `$dpm_cfg`

Variable name	Description
<code>\$enable</code>	Enable or disable dpm function.
<code>\$dpm_inline</code>	TRUE for inline DPM, FALSE for Modal DPM.
<code>\$grp_msk</code>	Motion group mask bit that can be used for DPM control. Allowed maximum of two active bits. Require re-power controller for change to be effective.
<code>\$bef_jbf</code>	TRUE: offset is applied before filter for inline DPM application, FALSE: offset is applied after filter for modal DPM application. Require re-cycle the power of the controller for the change to be effective.
<code>\$delay</code> (inline DPM use)	ITP counter that specifies window time that DPM will accept the offset after segment starts.
<code>\$ori_ctl</code>	True: Offset contains orientation control, False: w/o orientation control.
<code>\$cur_sch</code>	Schedule number in use
<code>\$max_sch</code>	n/a
<code>\$comp_sw</code>	Internal use
<code>\$debug</code>	Internal use

6.2 `$dpm_sch[sch_no]`

<code>\$comment</code>	Comment for the schedule
<code>\$dpm_on</code>	Enable or disable DPM for this schedule

6.3 \$dpm_sch[sch_no].\$grp[g]

\$dpm_type	1: for inline DPM, 2: for Modal DPM.
\$ofs_fram	1: PATH, 2: TOOLPATH, 3: UFRAME, 4:UTOOL.
\$ofs_accu	Model DPM - True: offset = new offset + un-consumed offset. - FALSE: offset = new offset. INLINE DPM - TRUE: The offset stored in position register will not be flushed after being used. - FALSE for Non-Accumulated: The offset stored in position register will be flushed after being used.
\$ofs_input (inline DPM use)	- USER-DEF: User defined offset. - SINGLE-PR: Offset stored at specified position register for all DPM move. - MULTI-PR: Offset stored at different position register where the register index must match the DPM move line number.
\$preg_idx (inline DPM use)	Position Register index for SINGLE-PR mode.
\$ofs_carry	n/a
\$ctr_rate	n/a
\$int_line (Inline DPM use)	Current interpolated motion line number updated at beginning of the dpm motion line.
\$ofs_line (Inline DPM use)	Line number when offset is applying to.
\$dat_rdy (Inline DPM use)	0: ready for offset input (set by DPM) 1: offset is ready (set by application). This variable is used to handshake to the external sensor signal.
\$ofs_stat (Inline DPM use)	This value reflects the offset application status: 1 offset is used successfully. 2 offset is ignored (beyond window time). 3 offset is ignored (done interpolating). 4 receive no offset at all.
\$snd_type	n/a
\$trk_mode (modal dpm use)	0: Normal, 1: stationary track. Used for Modal DPM FINE term type. At stationary track, robot is held at current motion line and continues responding to the path modification until the DI[sync_di] = ON.
\$sync_di (modal dpm use)	DI[sync_di] = OFF, continue stationary track until DI[sync_di] = ON.

6.4 \$dpm_sch[sch_no].\$grp[g].\$ofs[chn]

\$enable	Enable or disable offset channel.
\$otf_di_idx	OTF DIN index for OTF enable or disable DPM motion.
\$chn_type	Offset channel input type: 1: AI_TYPE, analog input. 2:GI_TYPE, group DI input. 3:BI_TYPE, bump digital input. 4: SV_TYPE, direct system variable input.
\$ramp_gain	n/a
\$scan_rate	n/a

\$ini_ofs	Channel offset set by user application with respect to offset reference frame. DPM reset it to zero after offset is read.
\$rem_ofs	Remaining channel offset to be applied.
\$app_ofs	Total applied offset value with respect to robot base.
\$in_tick	n/a
\$out_tick	n/a
\$stat	n/a
\$max_lim	Limit for maximum allowed total offset for the path.
\$min_lim	n/a
\$max_inc	Limit for the maximum allowed offset for each ITP control.
\$min_inc	Minimum threshold for the absolute offset to be considered.
\$a1	Constant for 1 st Exponential filter, default value 0.5.
\$a2	Constant for 2 nd Exponential filter, default value 0.5.

6.5 \$dpm_sch[sch_no].\$grp[g].\$ai_ch[chn]

\$p_gain	Positional control gain for the PID control.
\$d_gain	n/a
\$i_gain	n/a
\$str_cnt	Start to control count in ITP. DPM starts after cyc_num > \$str_cnt.
\$skp_cnt	First number of feedback samples to be skipped.
\$ave_cnt	The moving average sample numbers for the analog input signal.
\$fbk_mode	1: for MASTER mode 2: for feedback mode 3: for feedback1 mode.
\$mst_done	System init it FALSE at beginning of the track and sets it TRUE after reference value is calculated.
\$ref_val	Reference value used for the feedback control.
\$raw_fbk	Averaged analog input value for current cycle control.
\$cal_fbk	Calibrated analog input value for current cycle control.
\$port_num	Analog input port number.
\$cal_done	Set TRUE when calibration is done.
\$slope	Analog A/D calibration parameter.
\$intercept	Analog A/D calibration parameter.
\$cur_tick	Ros Tick for current control cycle.
\$lead_dis	n/a
\$buf_size	Ring buffer size must be larger than \$ave_num.
\$buf_cnt	Counter pointing to current feedback data.
\$t_adj	n/a
\$min_val	Minimum calibrated feedback, invalid data if data < \$min_val.
\$max_val	Maximum calibrated feedback, invalid data if data > \$max_val.
\$ctl_stage	Value for adaptive stage control. When > 0 enables the control.
\$parm1_f	Unit stage control value. Always > 0.
\$parm2_f	Max incremental threshold for stage control.
\$parm1_i	n/a
\$parm2_i	n/a

6.6 \$dpm_sch[sch_no].\$grp[g].\$bi_ch[chn]

\$di_idx1	Index for bump DIN for plus direction.
\$di_idx2	Index for bump DIN for minus direction.
\$bump_ofs	Bump offset when bump DIN is on.
\$bump_rate	Bump rate.
\$bump_gain	n/a

6.7 \$dpm_sch[sch_no].\$grp[g].\$gi_ch[chn]

\$di_idx	index for the group input .
\$scale	Scale factor for the group input.

7. DPM Programming

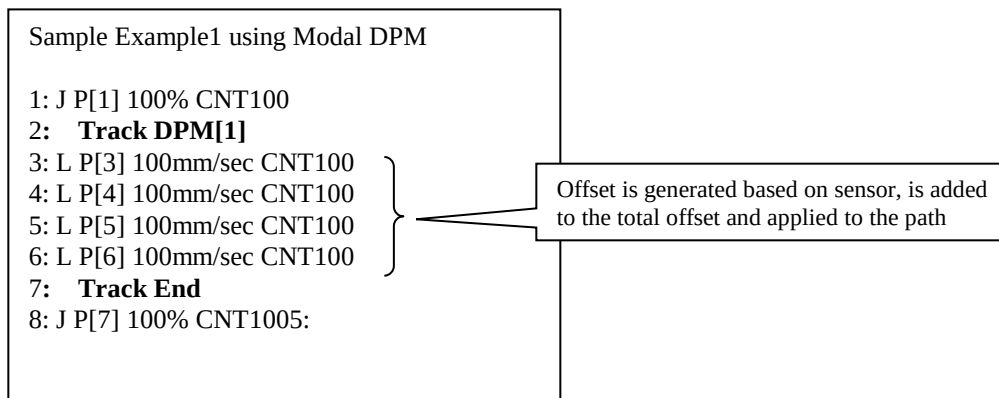
DPM supports modal and inline program instructions. Modal DPM is designed for the application that requires real time path modification based on sensor information along the entire motion path. The inline DPM is designed for the application that requires real time modification on destination position for each motion segment.

7.1 Modal DPM Programming

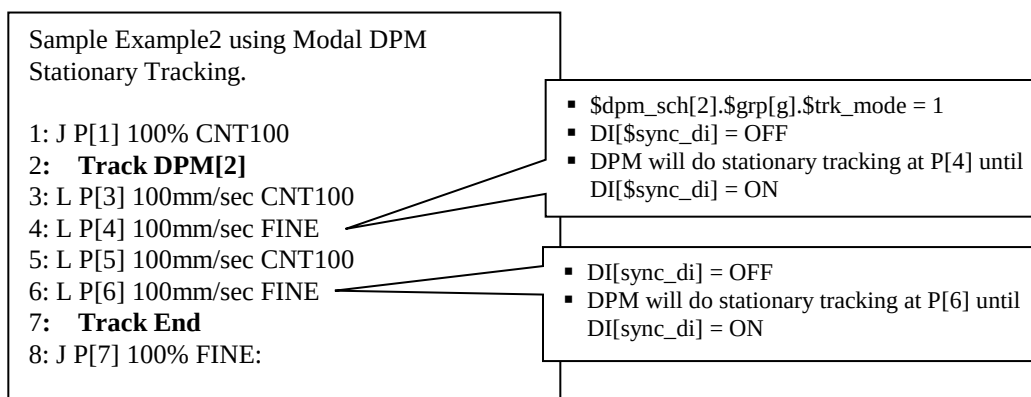
7.1.1 DPM path modification example

Modal DPM is for continuous path modification based on the sensor's input. It provides following two instructions

- Track DPM(sch)
- Track End

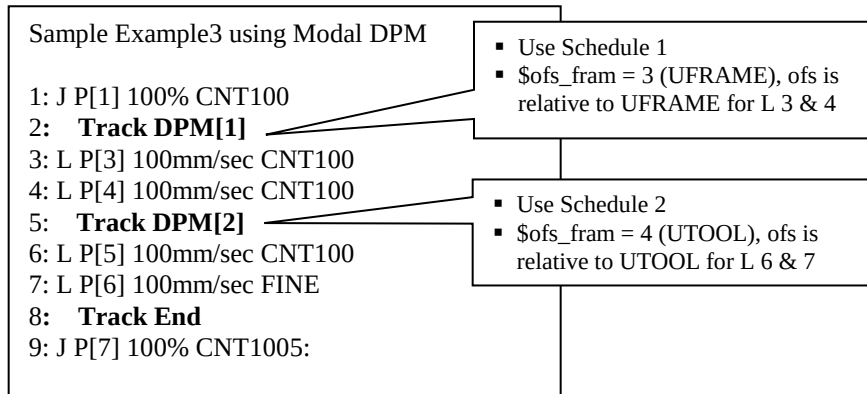


7.1.2 DPM Stationary Tracking example



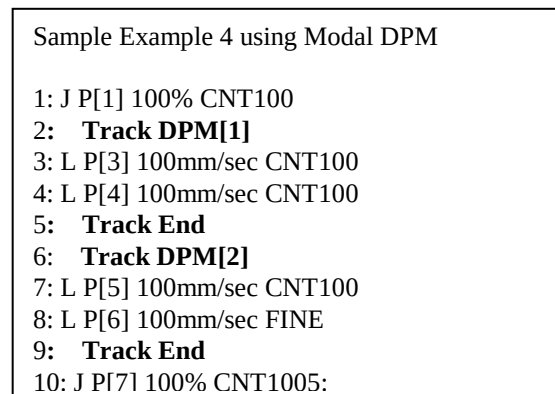
7.1.3 DPM Track to Track changing track schedules before Track End

In this case, the DPM track schedule changes before Track End. This allows user to change control parameters to different motion segments.



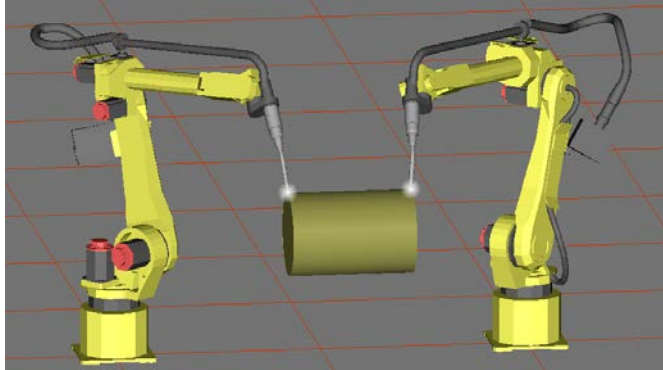
7.1.4 DPM Track to Track changing schedules after Track End

In this case, the DPM track schedule changes after Track End. There is no non-track motion point between Track End and Track DPM[2]. This also allows user to change control parameters to different motion segments.



7.1.5 Modal DPM Programming (Multiple Tasking and Stationary Track)

This example illustrates two arc welding robots welding a round tank. The two welding robots perform the independent stationary DPM track to a rotation tank.



Set stationary track for group 1 & 2 in DPM schedule

- `$dpm_sch[no].$grp[g].$strk_mode = 1`
- `$dpm_sch[no].$grp[g].$sync_di`

```

DEFAULT_GROUP = *,*,*,*,*,*,*,*
1: DO[11]=OFF ;
2: DO[12]=OFF ;
3: DO[10]=OFF ;
4: DO[13]=OFF ;
5: RUN GET_OFS ;
6: RUN G3_SUB ;
7: RUN DPM_TRC_G2 ;
8: RUN DPM_TRC_G1 ;

```

```
$dpm_sch[1].$grp[1].$strk_mode = 1
Set to stationary DPM until $DI[idx] = ON
idx = $dpm_sch[1].$grp[1].$sync_idx
```

```

/PROG G3_SUB
DEFAULT_GROUP      = *,*,1,*,*,*,*;;
1:J P[1] 100% FINE  ;
2: WAIT DO[10]=ON   ;
3: WAIT DO[11]=ON   ;
4: DO[12]=ON ;
5:J P[2] 60.0sec FINE  ;
6: DO[13]=ON ;

```

```
$dpm_sch[1].$grp[2].$strk_mode = 1
Set to stationary DPM until $DI[idx] = ON
idx = $dpm_sch[1].$grp[2].$sync_idx
```

```

/PROG DPM_TRC_G1
DEFAULT_GROUP      = 1,*,*,*,*,*,*;*
1:J P[1] 100% FINE  ;
2: WAIT 2.00(sec) ;
3:J P[2] 100% FINE  ;
4: DO[10]=ON ;
5: WAIT DO[12]=ON  ;
6: Arc Start E1[1] ;
7: Track DPM[1] ;
8:L P[3] 100mm/sec FINE  ;
9: Track End ;
10: Arc End E1[1] ;

```

```

/PROG DPM_TRC_G2
DEFAULT_GROUP      = *,1,*,*,*,*,*;*;
1:J P[1] 100% FINE  ;
2: WAIT 2.00(sec) ;
3:J P[2] 100% FINE  ;
4: DO[11]=ON ;
5: WAIT DO[12]=ON  ;
6: Arc Start E2[1] ;
7: Track DPM[1] ;
8:L P[3] 100mm/sec FINE  ;
9: Track End ;
10: Arc End E2[1] ;

```

7.2 Inline DPM Programming

7.2.1 Single Group Inline DPM Programming

```
Single Group DPM motion [1,*,*,*,*,*,*]  
$dpm_cfg.$grp_msk = 1  
  
1: J P[1] 100% CNT100  
2: L P[2] 100mm/sec CNT1 DPM[1]  
3: L P[3] 100mm/sec CNT10 DPM[1]  
4: L P[4] 100mm/sec CNT100 DPM[1]  
5: J P[5] 100% CNT100  
6: L P[6] 100mm/sec FINE DPM[1]  
7: J P[7] 100% CNT1005:  
8: L P[8] 100mm/sec FINE DPM[1]  
9: J P[9] 100% CNT100
```

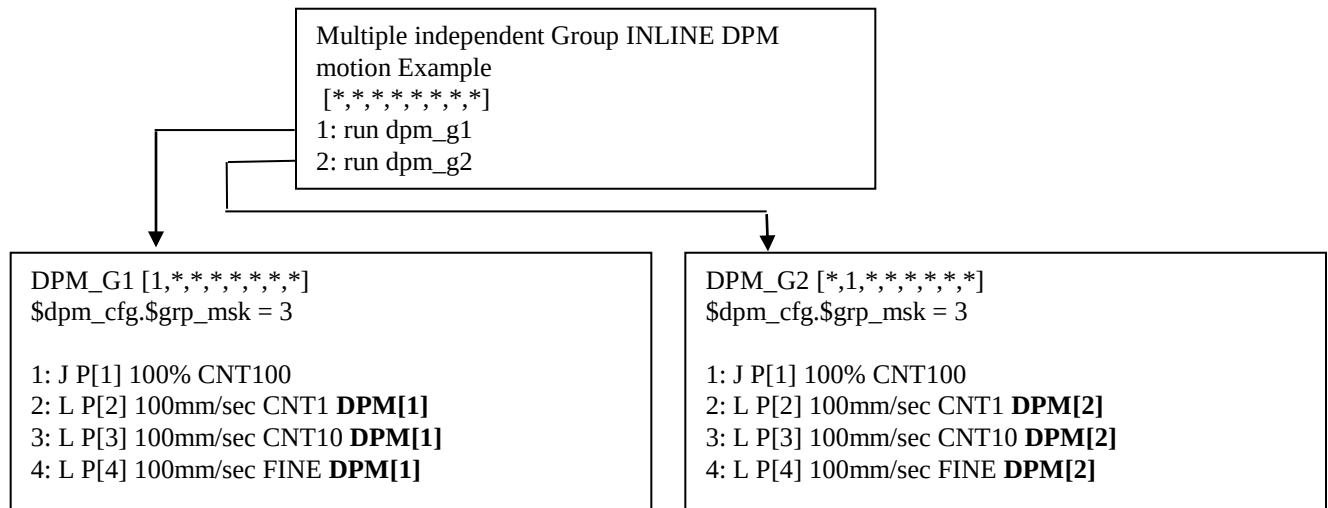
7.2.2 Simultaneous Multiple Group INLINE DPM motion

DPM supports simultaneous group DPM motion. All the robot group motion starts and stops at the same time and each robot group modify its destination based on its own offset input.

```
Multiple Group DPM motion [1,1,*,*,*,*,*]  
$dpm_cfg.$grp_msk = 3  
  
1: J P[1] 100% CNT100  
2: L P[2] 100mm/sec CNT5 DPM[1]  
3: L P[3] 100mm/sec CNT100 DPM[1]  
4: L P[4] 100mm/sec FINE DPM[1]
```

7.2.3 Multiple Independent Group Inline DPM Programming

In this case, robot group motion is not synchronized and each robot group modifies its destination based on its own offset input.



8. Error Messages

Alarm	Severity	Description
DPM-002 Cyc Offset Limit Err	Stop	interpolated offset exceeds the allowed bound (\$dpm_sch[.grp[.].\$max_cyc)
DPM-003 Offset is ignored	Warning	current segment is done interpolating
DPM-004 Offset is too late	Warning	Offset is available after window time
DPM-005 No Zero Dist Move	Stop	Zero distance DPM motion is not allowed
DPM-006 No Joint motion	Stop	Attempt to offset a JOINT motion
DPM-013 NO Sysvar loaded	Stop	DPM is not installed properly
DPM-014 Dest Offset Limit Error	Stop	Destination offset exceeds the allowed bound (\$dpm_sch[.grp[.].\$max_lim)
DPM-015 Analog IN not set	Stop	Analog input is not set properly
DPM-016 Group IN not set	Stop	Group input is not set properly
DPM-017 Bump IN not set	Stop	Bump digital input is not set properly
DPM-018 Analog IN not calibrated	Stop	Analog input is not calibrated
DPM-019 Group IN not calibrated	Stop	Need set none zero scale factor for group Input.
DPM-020 Inline DPM is not set	Stop	Need set \$dpm_cfg.\$enable = TRUE
DPM-021 WJNT is not supported	Stop	Attempt to offset a Wrist Joint motion
DPM-022 UTOOL is changed	Stop	Attempt to OTF changing an UTOOL
DPM-023 Wrong Schedule Number	Stop	Wrong schedule number
DPM-024 Can't control orientation	Stop	DPM orientation is prohibited at non mode3 resume. Set \$dpm_cfg.\$ori_ctl = FALSE.
DPM-025 Can't enb ori_ctl for PATH	Stop	DPM orientation is NOT supported for TPATH and PATH. Use UFRAME or UTOOL offset relative for orientation control.
DPM-026 Fail to create offset frame	Warning	Invalid robot move direction. Use UFRAME or UTOOL offset frame reference.
DPM-027 Mismatch DPM type	Stop	DPM type mismatch in configuration and schedule.
DPM-028 Buffer is too short	Abort	Buffer size is short to hold the data for analog input. Increase the buffer size and repower the controller.